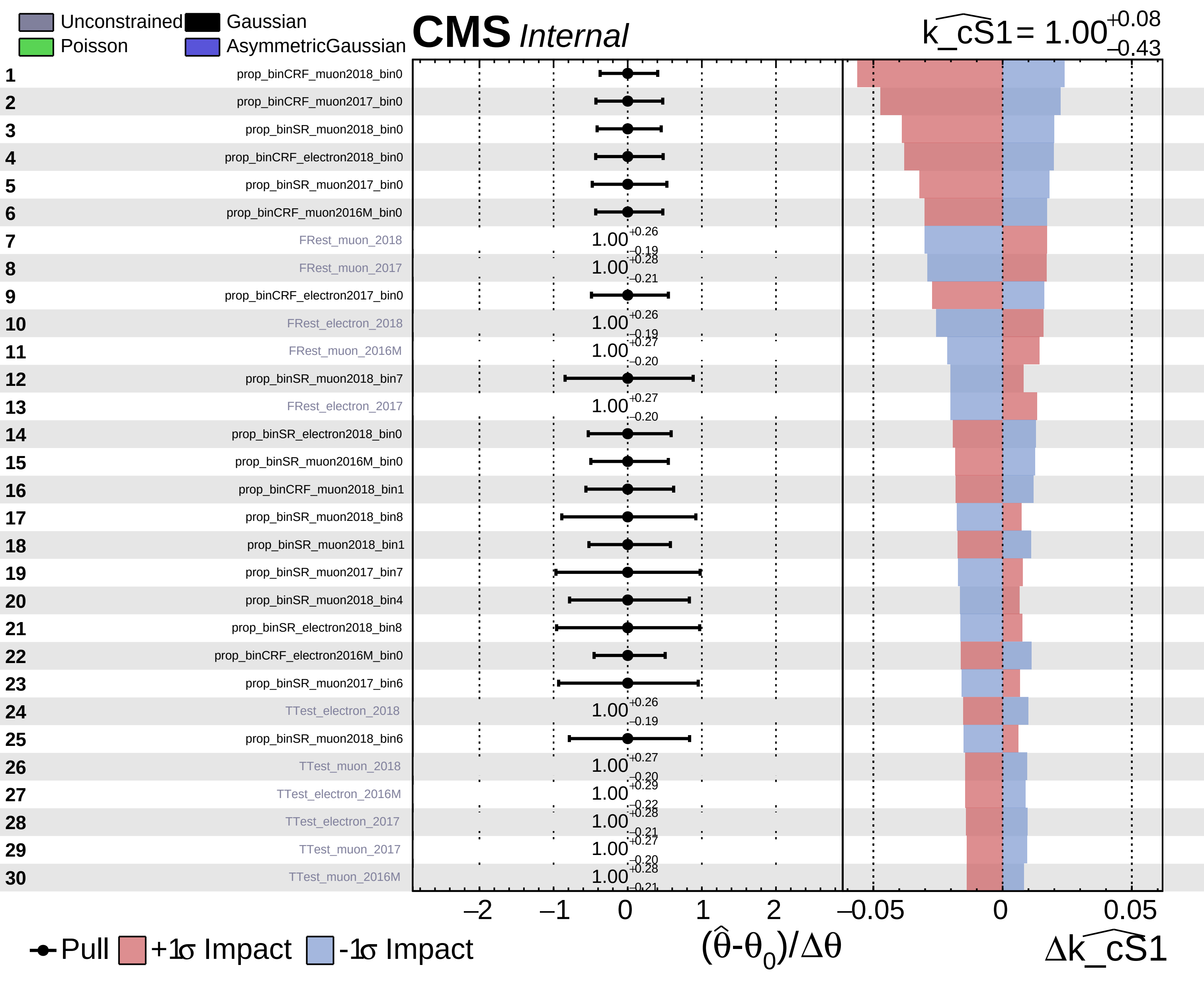
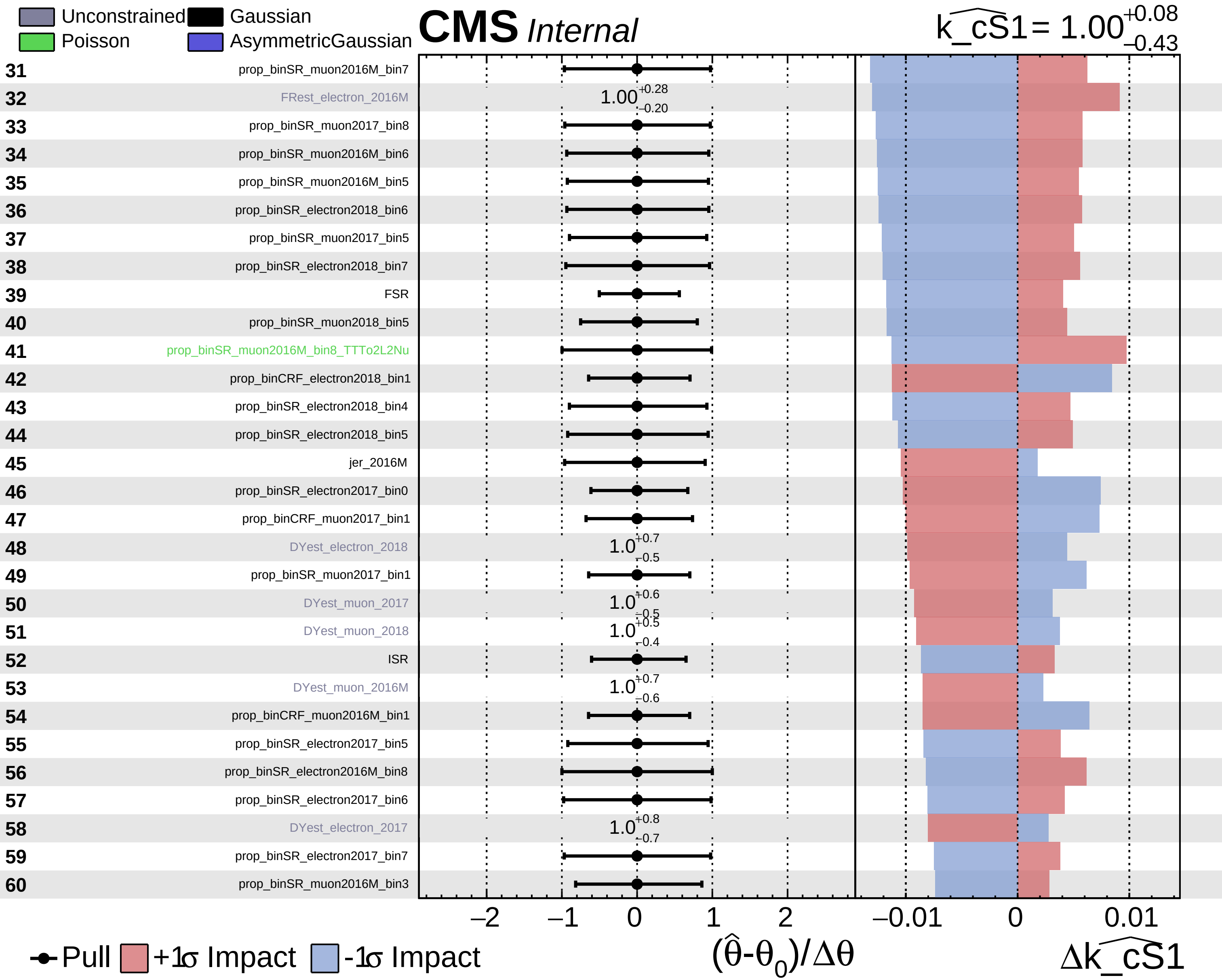
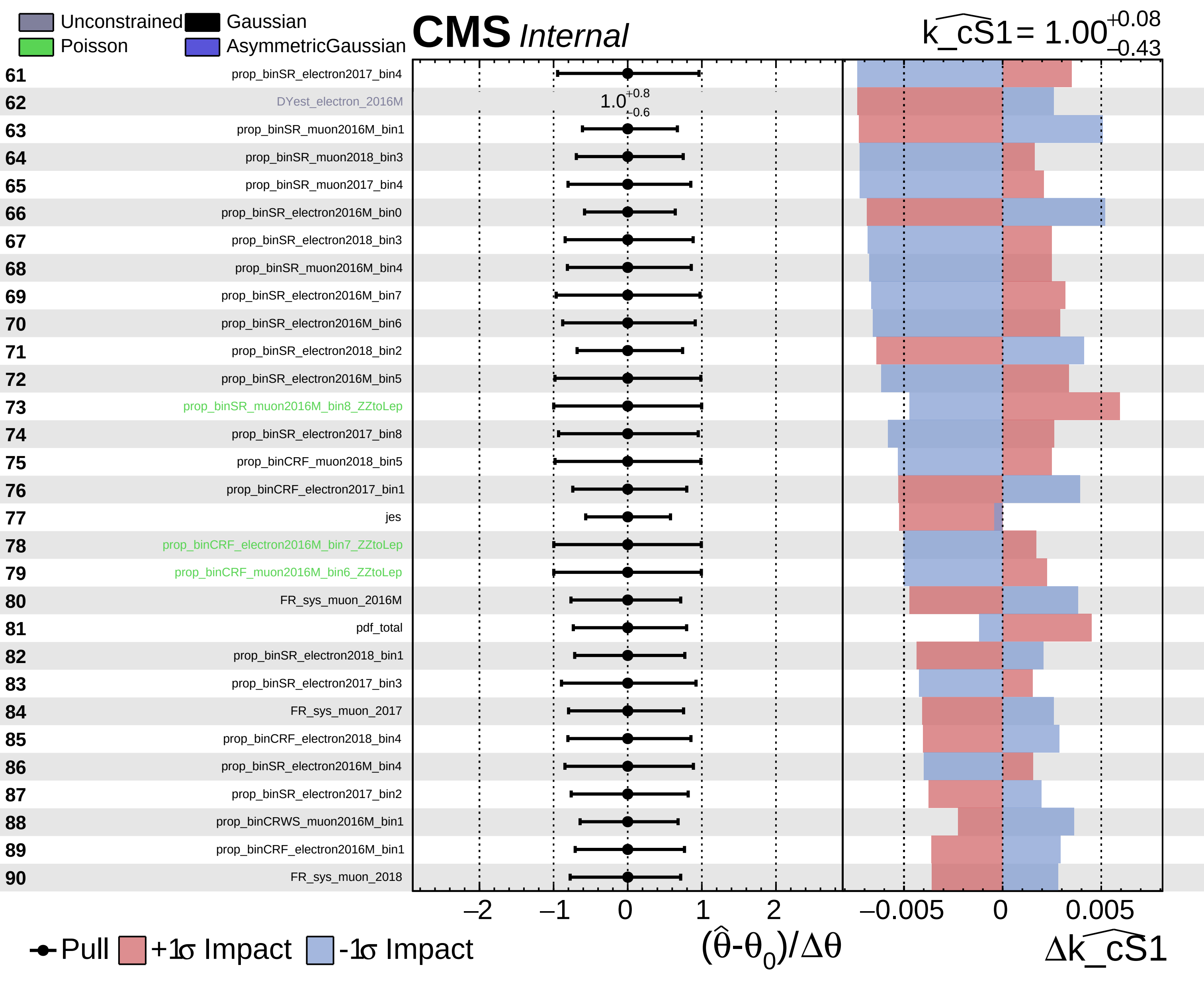
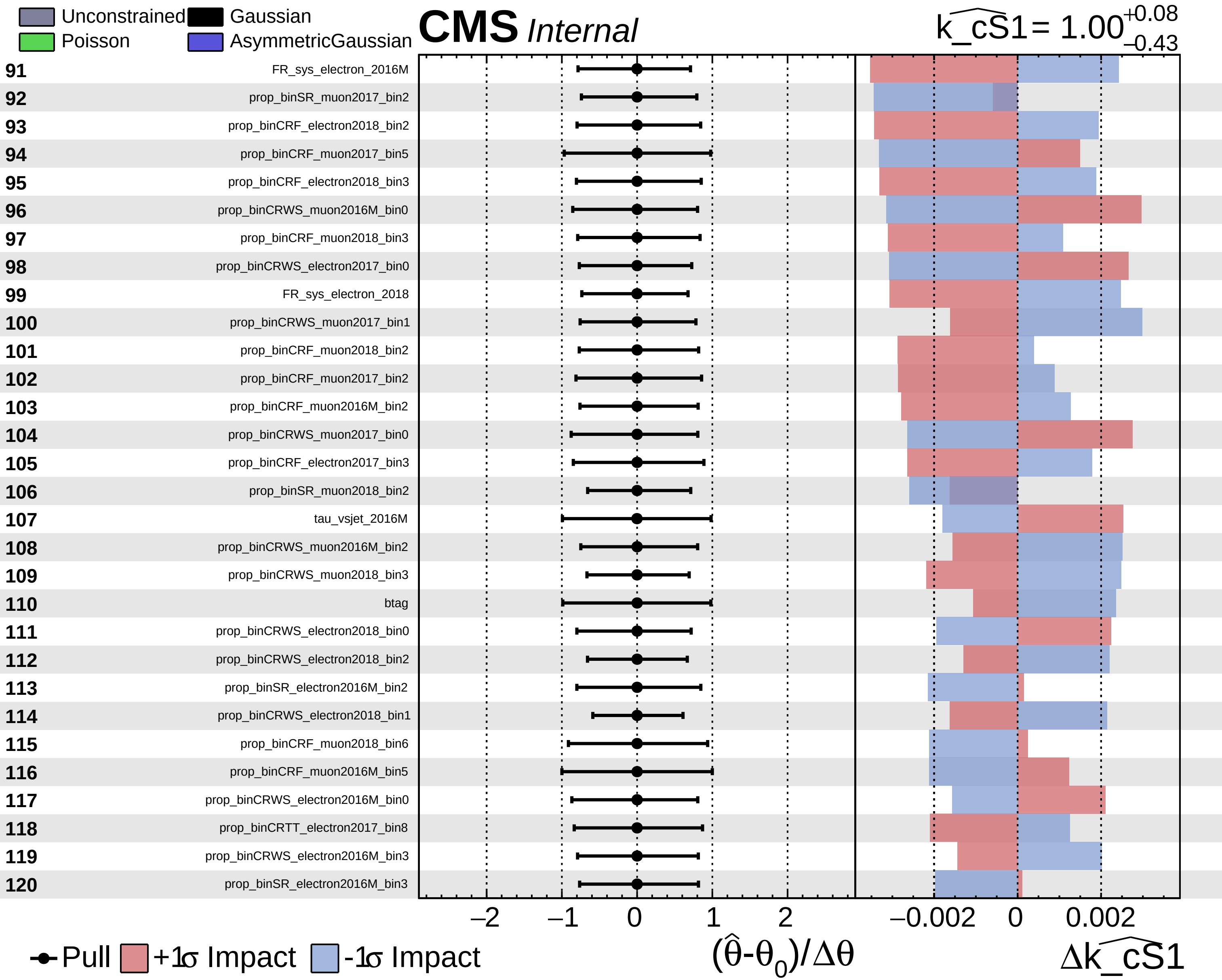

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS1 = 1.00 +0.08 −0.43
1 prop_binCRF_muon2018_bin0
2 prop_binCRF_muon2017_bin0
3 prop_binSR_muon2018_bin0
4 prop_binCRF_electron2018_bin0
5 prop_binSR_muon2017_bin0
6 prop_binCRF_muon2016M_bin0
7 FRest_muon_2018 1.00 +0.26 −0.19
8 FRest_muon_2017 1.00 +0.28 −0.21
9 prop_binCRF_electron2017_bin0
10 FRest_electron_2018 1.00 +0.26 −0.19
11 FRest_muon_2016M 1.00 +0.27 −0.20
12 prop_binSR_muon2018_bin7
13 FRest_electron_2017 1.00 +0.27 −0.20
14 prop_binSR_electron2018_bin0
15 prop_binSR_muon2016M_bin0
16 prop_binCRF_muon2018_bin1
17 prop_binSR_muon2018_bin8
18 prop_binSR_muon2018_bin1
19 prop_binSR_muon2017_bin7
20 prop_binSR_muon2018_bin4
21 prop_binSR_electron2018_bin8
22 prop_binCRF_electron2016M_bin0
23 prop_binSR_muon2017_bin6
24 TTest_electron_2018 1.00 +0.26 −0.19
25 prop_binSR_muon2018_bin6
26 TTest_muon_2018 1.00 +0.27 −0.20
27 TTest_electron_2016M 1.00 +0.29 −0.22
28 TTest_electron_2017 1.00 +0.28 −0.21
29 TTest_muon_2017 1.00 +0.27 −0.20
30 TTest_muon_2016M 1.00 +0.28 −0.21
−2 −1 0 1 2
−0.05 0 0.05
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cS1



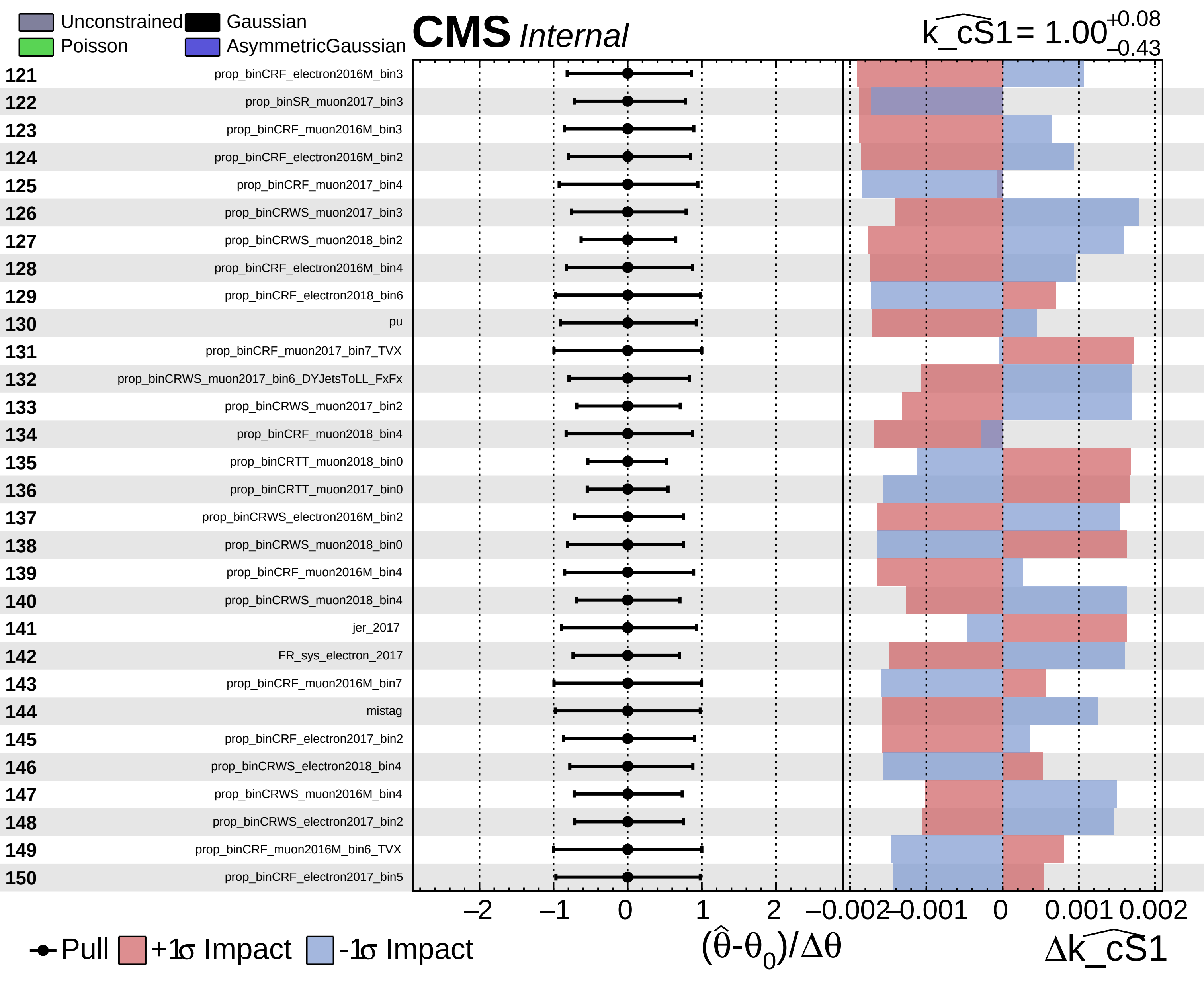
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS1 = 1.00 +0.08 −0.43
31 prop_binSR_muon2016M_bin7
32 FRest_electron_2016M 1.00 +0.28 −0.20
33 prop_binSR_muon2017_bin8
34 prop_binSR_muon2016M_bin6
35 prop_binSR_muon2016M_bin5
36 prop_binSR_electron2018_bin6
37 prop_binSR_muon2017_bin5
38 prop_binSR_electron2018_bin7
39 FSR
40 prop_binSR_muon2018_bin5
41 prop_binSR_muon2016M_bin8_TTTo2L2Nu
42 prop_binCRF_electron2018_bin1
43 prop_binSR_electron2018_bin4
44 prop_binSR_electron2018_bin5
45 jer_2016M
46 prop_binSR_electron2017_bin0
47 prop_binCRF_muon2017_bin1
48 DYest_electron_2018 1.0 +0.7 −0.5
49 prop_binSR_muon2017_bin1
50 DYest_muon_2017 1.0 +0.6 −0.5
51 DYest_muon_2018 1.0 +0.5 −0.4
52 ISR
53 DYest_muon_2016M 1.0 +0.7 −0.6
54 prop_binCRF_muon2016M_bin1
55 prop_binSR_electron2017_bin5
56 prop_binSR_electron2016M_bin8
57 prop_binSR_electron2017_bin6
58 DYest_electron_2017 1.0 +0.8 −0.7
59 prop_binSR_electron2017_bin7
60 prop_binSR_muon2016M_bin3
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cS1



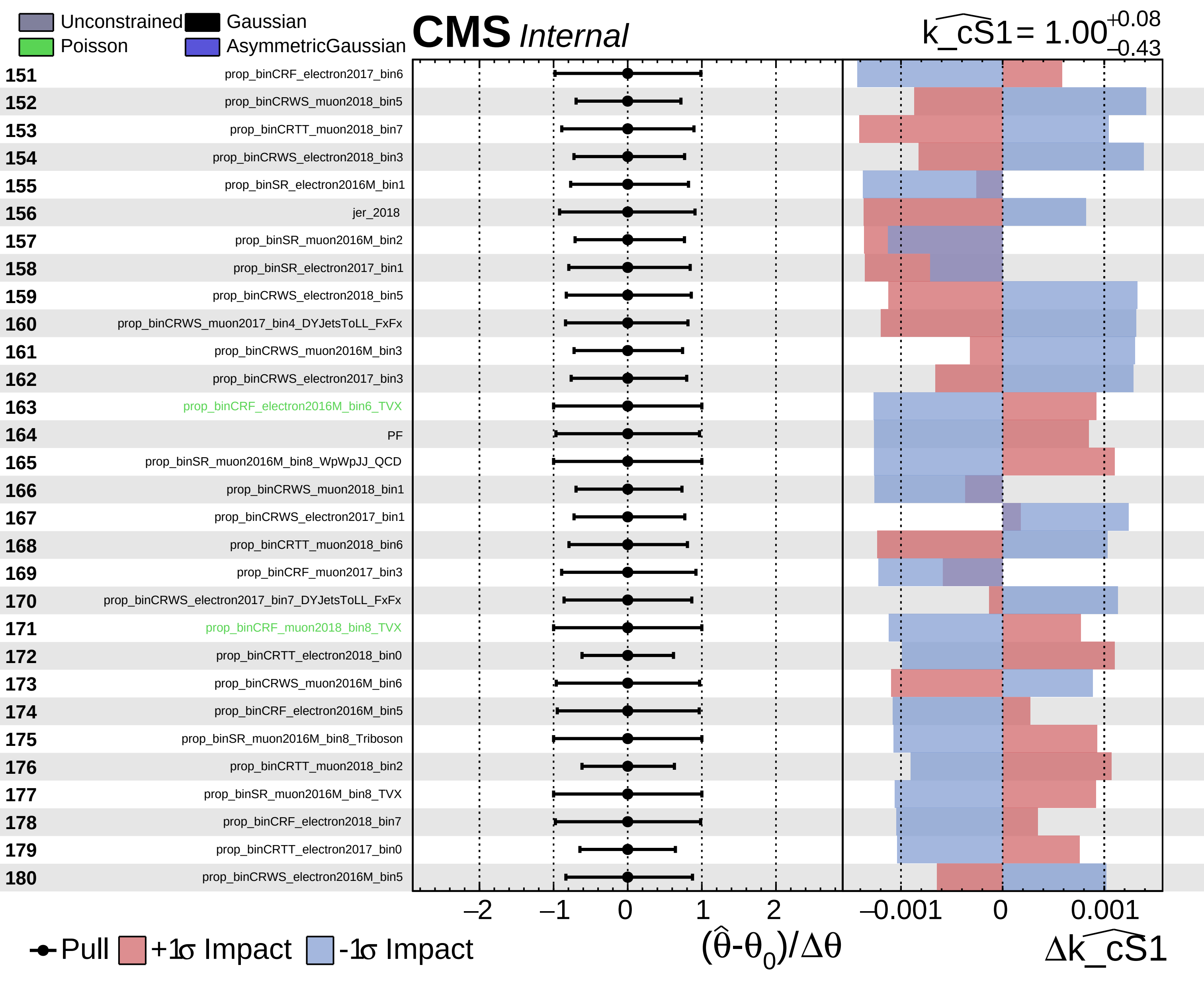
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS1 = 1.00 +0.08 −0.43
61 prop_binSR_electron2017_bin4
62 DYest_electron_2016M
1.0 +0.8 −0.6
63 prop_binSR_muon2016M_bin1
64 prop_binSR_muon2018_bin3
65 prop_binSR_muon2017_bin4
66 prop_binSR_electron2016M_bin0
67 prop_binSR_electron2018_bin3
68 prop_binSR_muon2016M_bin4
69 prop_binSR_electron2016M_bin7
70 prop_binSR_electron2016M_bin6
71 prop_binSR_electron2018_bin2
72 prop_binSR_electron2016M_bin5
73 prop_binSR_muon2016M_bin8_ZZtoLep
74 prop_binSR_electron2017_bin8
75 prop_binCRF_muon2018_bin5
76 prop_binCRF_electron2017_bin1
77 jes
78 prop_binCRF_electron2016M_bin7_ZZtoLep
79 prop_binCRF_muon2016M_bin6_ZZtoLep
80 FR_sys_muon_2016M
81 pdf_total
82 prop_binSR_electron2018_bin1
83 prop_binSR_electron2017_bin3
84 FR_sys_muon_2017
85 prop_binCRF_electron2018_bin4
86 prop_binSR_electron2016M_bin4
87 prop_binSR_electron2017_bin2
88 prop_binCRWS_muon2016M_bin1
89 prop_binCRF_electron2016M_bin1
90 FR_sys_muon_2018
−2 −1 0 1 2
−0.005 0 0.005
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cS1



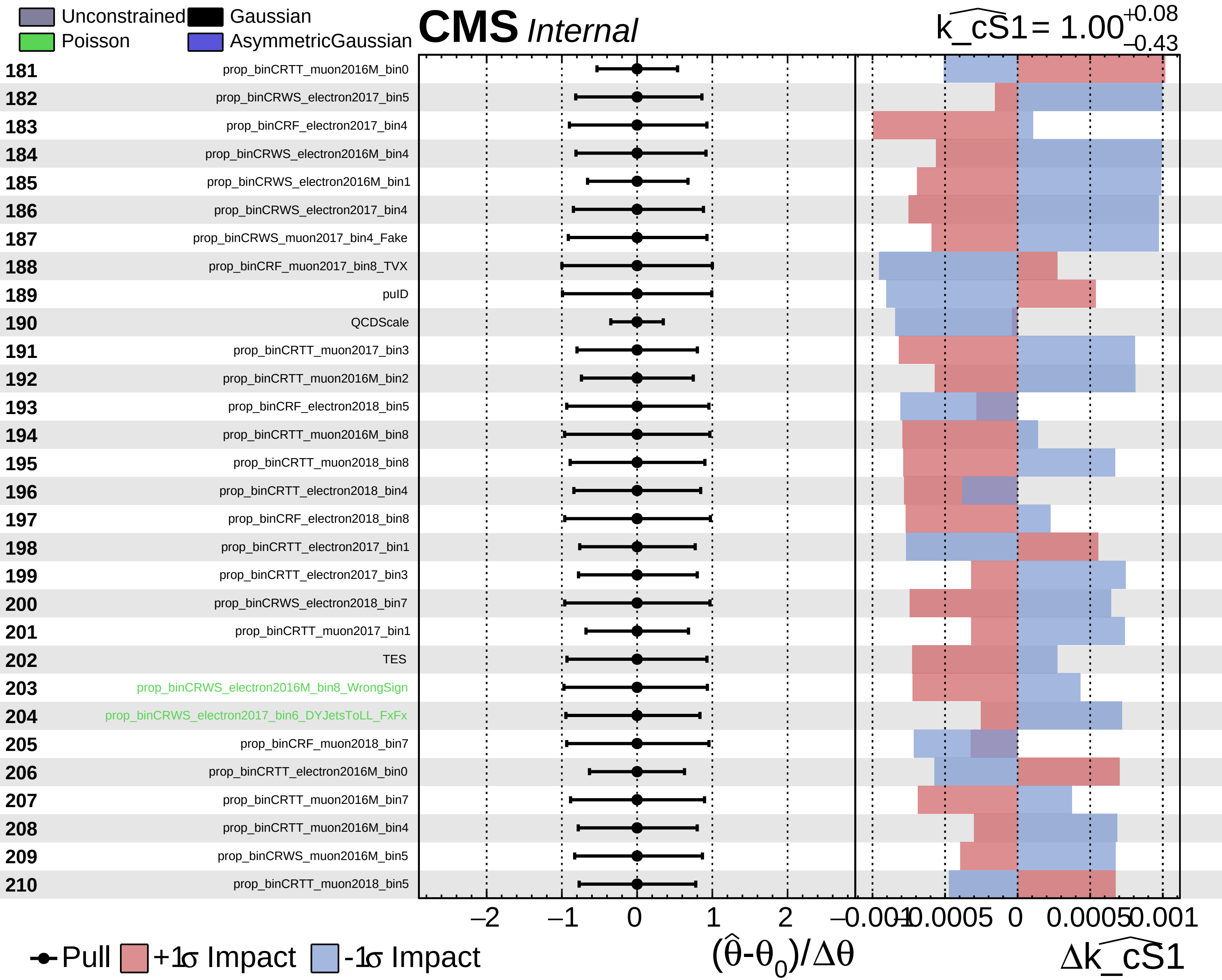
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS1 = 1.00 +0.08 −0.43
91 FR_sys_electron_2016M
92 prop_binSR_muon2017_bin2
93 prop_binCRF_electron2018_bin2
94 prop_binCRF_muon2017_bin5
95 prop_binCRF_electron2018_bin3
96 prop_binCRWS_muon2016M_bin0
97 prop_binCRF_muon2018_bin3
98 prop_binCRWS_electron2017_bin0
99 FR_sys_electron_2018
100 prop_binCRWS_muon2017_bin1
101 prop_binCRF_muon2018_bin2
102 prop_binCRF_muon2017_bin2
103 prop_binCRF_muon2016M_bin2
104 prop_binCRWS_muon2017_bin0
105 prop_binCRF_electron2017_bin3
106 prop_binSR_muon2018_bin2
107 tau_vsjet_2016M
108 prop_binCRWS_muon2016M_bin2
109 prop_binCRWS_muon2018_bin3
110 btag
111 prop_binCRWS_electron2018_bin0
112 prop_binCRWS_electron2018_bin2
113 prop_binSR_electron2016M_bin2
114 prop_binCRWS_electron2018_bin1
115 prop_binCRF_muon2018_bin6
116 prop_binCRF_muon2016M_bin5
117 prop_binCRWS_electron2016M_bin0
118 prop_binCRTT_electron2017_bin8
119 prop_binCRWS_electron2016M_bin3
120 prop_binSR_electron2016M_bin3
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS1



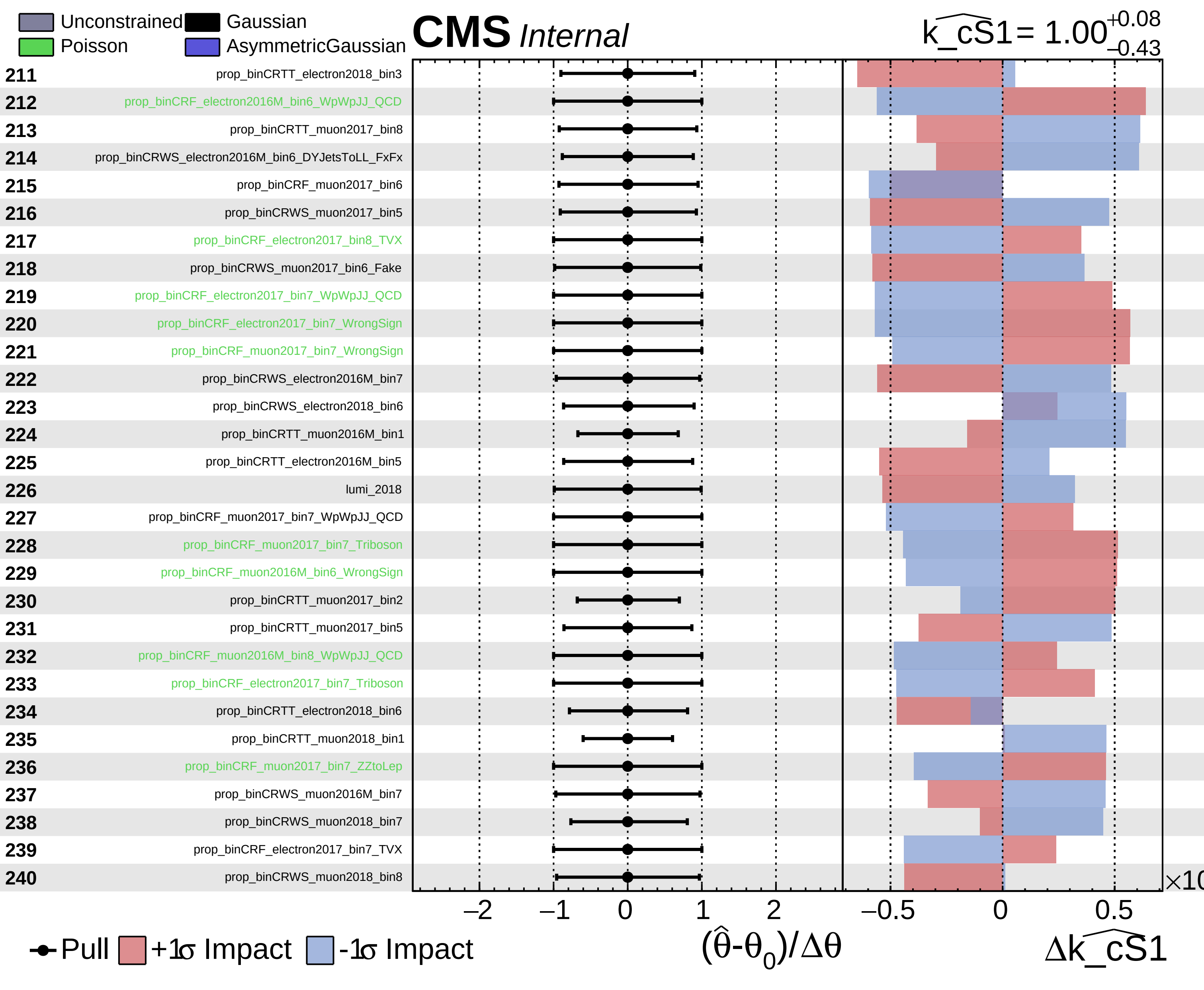
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS1 = 1.00 +0.08 −0.43
121 prop_binCRF_electron2016M_bin3
122 prop_binSR_muon2017_bin3
123 prop_binCRF_muon2016M_bin3
124 prop_binCRF_electron2016M_bin2
125 prop_binCRF_muon2017_bin4
126 prop_binCRWS_muon2017_bin3
127 prop_binCRWS_muon2018_bin2
128 prop_binCRF_electron2016M_bin4
129 prop_binCRF_electron2018_bin6
130 pu
131 prop_binCRF_muon2017_bin7_TVX
132 prop_binCRWS_muon2017_bin6_DYJetsToLL_FxFx
133 prop_binCRWS_muon2017_bin2
134 prop_binCRF_muon2018_bin4
135 prop_binCRTT_muon2018_bin0
136 prop_binCRTT_muon2017_bin0
137 prop_binCRWS_electron2016M_bin2
138 prop_binCRWS_muon2018_bin0
139 prop_binCRF_muon2016M_bin4
140 prop_binCRWS_muon2018_bin4
141 jer_2017
142 FR_sys_electron_2017
143 prop_binCRF_muon2016M_bin7
144 mistag
145 prop_binCRF_electron2017_bin2
146 prop_binCRWS_electron2018_bin4
147 prop_binCRWS_muon2016M_bin4
148 prop_binCRWS_electron2017_bin2
149 prop_binCRF_muon2016M_bin6_TVX
150 prop_binCRF_electron2017_bin5
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cS1



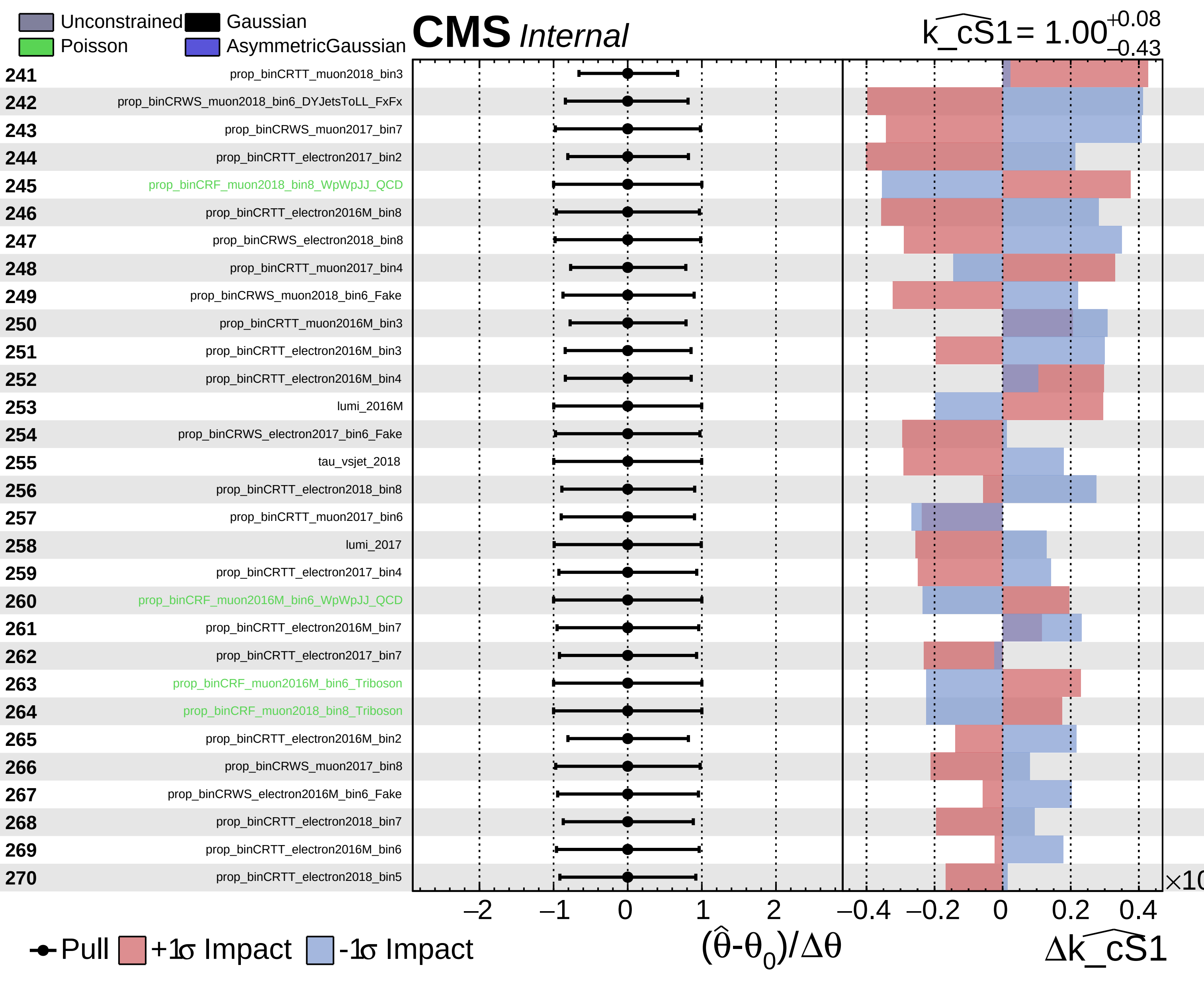
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS1} = 1.00^{+0.08}_{-0.43}$
151 prop_binCRF_electron2017_bin6
152 prop_binCRWS_muon2018_bin5
153 prop_binCRTT_muon2018_bin7
154 prop_binCRWS_electron2018_bin3
155 prop_binSR_electron2016M_bin1
156 jer_2018
157 prop_binSR_muon2016M_bin2
158 prop_binSR_electron2017_bin1
159 prop_binCRWS_electron2018_bin5
160 prop_binCRWS_muon2017_bin4_DYJetsToLL_FxFx
161 prop_binCRWS_muon2016M_bin3
162 prop_binCRWS_electron2017_bin3
163 prop_binCRF_electron2016M_bin6_TVX
164 PF
165 prop_binSR_muon2016M_bin8_WpWpJJ_QCD
166 prop_binCRWS_muon2018_bin1
167 prop_binCRWS_electron2017_bin1
168 prop_binCRTT_muon2018_bin6
169 prop_binCRF_muon2017_bin3
170 prop_binCRWS_electron2017_bin7_DYJetsToLL_FxFx
171 prop_binCRF_muon2018_bin8_TVX
172 prop_binCRTT_electron2018_bin0
173 prop_binCRWS_muon2016M_bin6
174 prop_binCRF_electron2016M_bin5
175 prop_binSR_muon2016M_bin8_Triboson
176 prop_binCRTT_muon2018_bin2
177 prop_binSR_muon2016M_bin8_TVX
178 prop_binCRF_electron2018_bin7
179 prop_binCRTT_electron2017_bin0
180 prop_binCRWS_electron2016M_bin5
−2 −1 0 1 2
−0.001 0 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS1}$



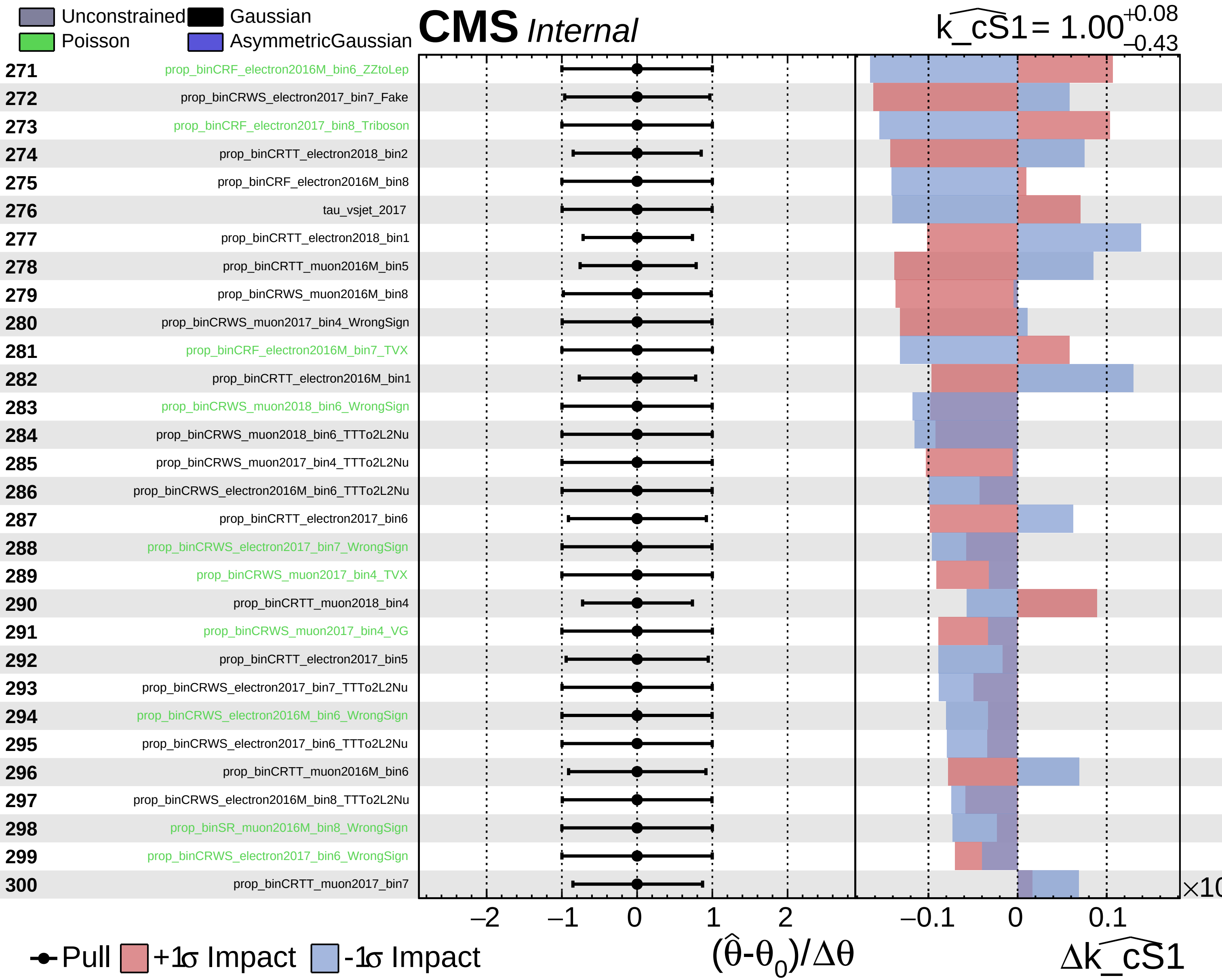
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS1 = 1.00 +0.08 −0.43
181 prop_binCRTT_muon2016M_bin0
182 prop_binCRWS_electron2017_bin5
183 prop_binCRF_electron2017_bin4
184 prop_binCRWS_electron2016M_bin4
185 prop_binCRWS_electron2016M_bin1
186 prop_binCRWS_electron2017_bin4
187 prop_binCRWS_muon2017_bin4_Fake
188 prop_binCRF_muon2017_bin8_TVX
189 puID
190 QCDScale
191 prop_binCRTT_muon2017_bin3
192 prop_binCRTT_muon2016M_bin2
193 prop_binCRF_electron2018_bin5
194 prop_binCRTT_muon2016M_bin8
195 prop_binCRTT_muon2018_bin8
196 prop_binCRTT_electron2018_bin4
197 prop_binCRF_electron2018_bin8
198 prop_binCRTT_electron2017_bin1
199 prop_binCRTT_electron2017_bin3
200 prop_binCRWS_electron2018_bin7
201 prop_binCRTT_muon2017_bin1
202 TES
203 prop_binCRWS_electron2016M_bin8_WrongSign
204 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
205 prop_binCRF_muon2018_bin7
206 prop_binCRTT_electron2016M_bin0
207 prop_binCRTT_muon2016M_bin7
208 prop_binCRTT_muon2016M_bin4
209 prop_binCRWS_muon2016M_bin5
210 prop_binCRTT_muon2018_bin5
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cS1



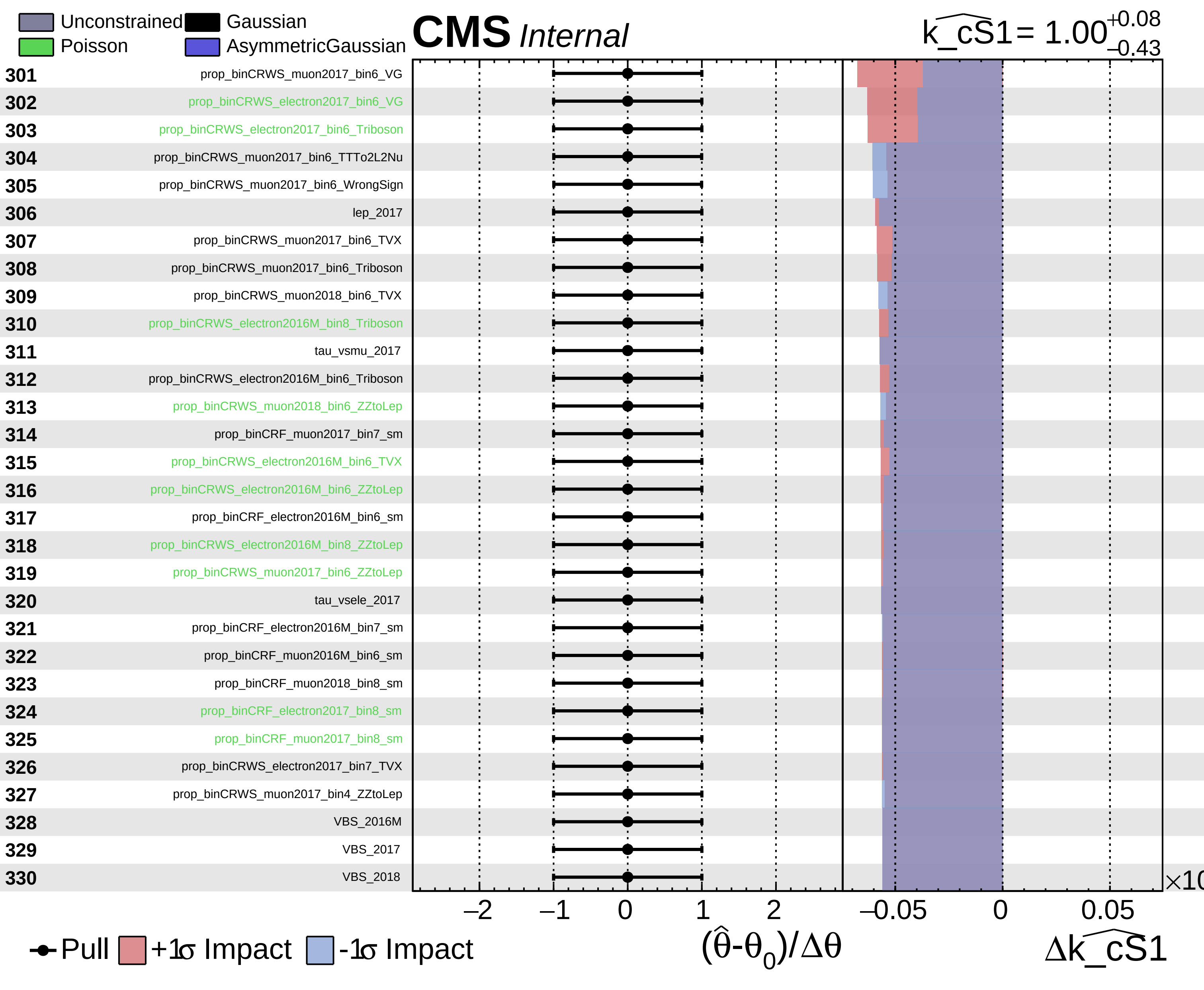
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS1} = 1.00^{+0.08}_{-0.43}$
211 prop_binCRTT_electron2018_bin3
212 prop_binCRF_electron2016M_bin6_WpWpJJ_QCD
213 prop_binCRTT_muon2017_bin8
214 prop_binCRWS_electron2016M_bin6_DYJetsToLL_FxFx
215 prop_binCRF_muon2017_bin6
216 prop_binCRWS_muon2017_bin5
217 prop_binCRF_electron2017_bin8_TVX
218 prop_binCRWS_muon2017_bin6_Fake
219 prop_binCRF_electron2017_bin7_WpWpJJ_QCD
220 prop_binCRF_electron2017_bin7_WrongSign
221 prop_binCRF_muon2017_bin7_WrongSign
222 prop_binCRWS_electron2016M_bin7
223 prop_binCRWS_electron2018_bin6
224 prop_binCRTT_muon2016M_bin1
225 prop_binCRTT_electron2016M_bin5
226 lumi_2018
227 prop_binCRF_muon2017_bin7_WpWpJJ_QCD
228 prop_binCRF_muon2017_bin7_Triboson
229 prop_binCRF_muon2016M_bin6_WrongSign
230 prop_binCRTT_muon2017_bin2
231 prop_binCRTT_muon2017_bin5
232 prop_binCRF_muon2016M_bin8_WpWpJJ_QCD
233 prop_binCRF_electron2017_bin7_Triboson
234 prop_binCRTT_electron2018_bin6
235 prop_binCRTT_muon2018_bin1
236 prop_binCRF_muon2017_bin7_ZZtoLep
237 prop_binCRWS_muon2016M_bin7
238 prop_binCRWS_muon2018_bin7
239 prop_binCRF_electron2017_bin7_TVX
240 prop_binCRWS_muon2018_bin8
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS1}$



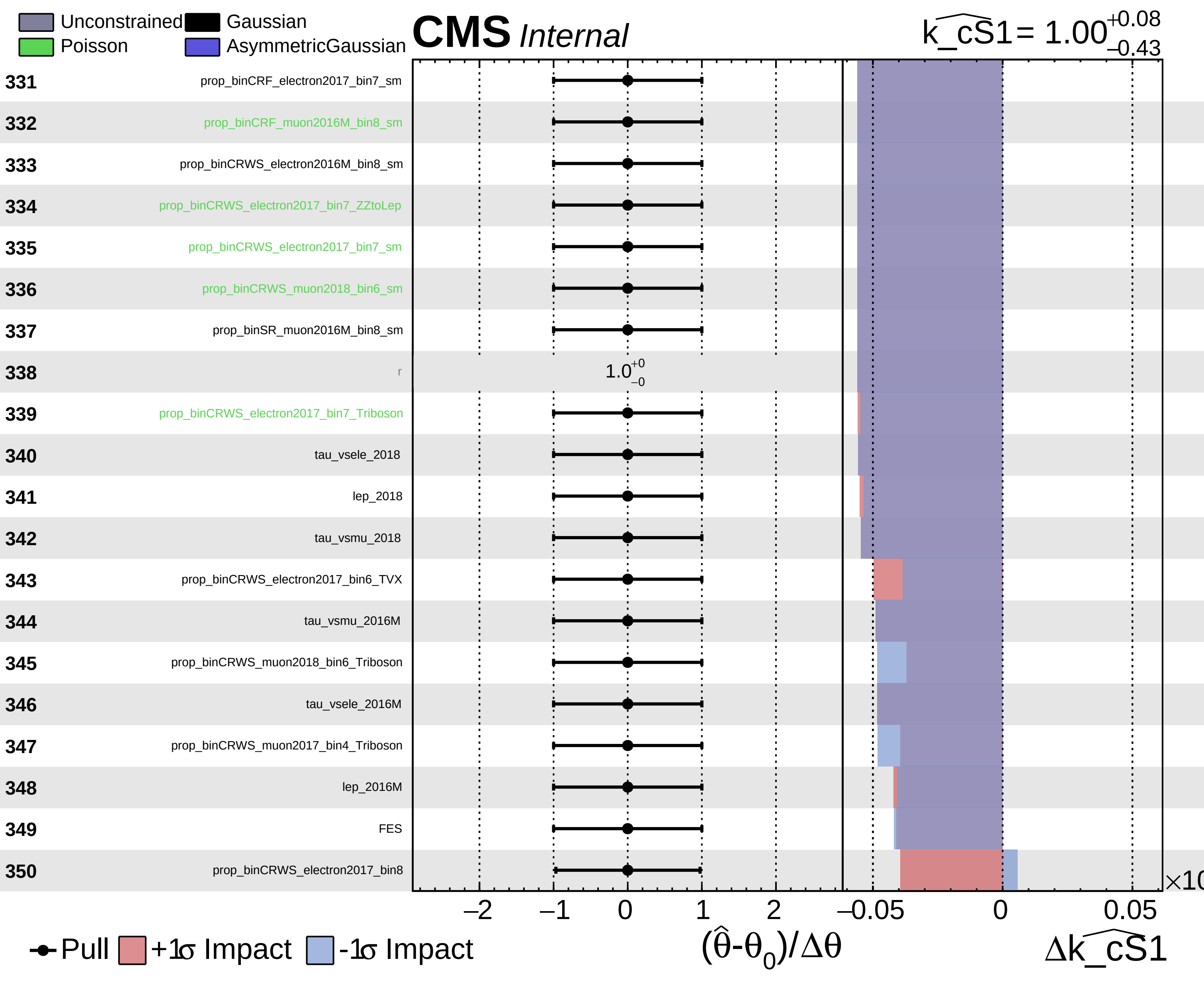
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS1} = 1.00^{+0.08}_{-0.43}$
241 prop_binCRTT_muon2018_bin3
242 prop_binCRWS_muon2018_bin6_DYJetsToLL_FxFx
243 prop_binCRWS_muon2017_bin7
244 prop_binCRTT_electron2017_bin2
245 prop_binCRF_muon2018_bin8_WpWpJJ_QCD
246 prop_binCRTT_electron2016M_bin8
247 prop_binCRWS_electron2018_bin8
248 prop_binCRTT_muon2017_bin4
249 prop_binCRWS_muon2018_bin6_Fake
250 prop_binCRTT_muon2016M_bin3
251 prop_binCRTT_electron2016M_bin3
252 prop_binCRTT_electron2016M_bin4
253 lumi_2016M
254 prop_binCRWS_electron2017_bin6_Fake
255 tau_vsjet_2018
256 prop_binCRTT_electron2018_bin8
257 prop_binCRTT_muon2017_bin6
258 lumi_2017
259 prop_binCRTT_electron2017_bin4
260 prop_binCRF_muon2016M_bin6_WpWpJJ_QCD
261 prop_binCRTT_electron2016M_bin7
262 prop_binCRTT_electron2017_bin7
263 prop_binCRF_muon2016M_bin6_Triboson
264 prop_binCRF_muon2018_bin8_Triboson
265 prop_binCRTT_electron2016M_bin2
266 prop_binCRWS_muon2017_bin8
267 prop_binCRWS_electron2016M_bin6_Fake
268 prop_binCRTT_electron2018_bin7
269 prop_binCRTT_electron2016M_bin6
270 prop_binCRTT_electron2018_bin5
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS1}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS1} = 1.00^{+0.08}_{-0.43}$
271 prop_binCRF_electron2016M_bin6_ZZtoLep
272 prop_binCRWS_electron2017_bin7_Fake
273 prop_binCRF_electron2017_bin8_Triboson
274 prop_binCRTT_electron2018_bin2
275 prop_binCRF_electron2016M_bin8
276 tau_vsjet_2017
277 prop_binCRTT_electron2018_bin1
278 prop_binCRTT_muon2016M_bin5
279 prop_binCRWS_muon2016M_bin8
280 prop_binCRWS_muon2017_bin4_WrongSign
281 prop_binCRF_electron2016M_bin7_TVX
282 prop_binCRTT_electron2016M_bin1
283 prop_binCRWS_muon2018_bin6_WrongSign
284 prop_binCRWS_muon2018_bin6_TTTo2L2Nu
285 prop_binCRWS_muon2017_bin4_TTTo2L2Nu
286 prop_binCRWS_electron2016M_bin6_TTTo2L2Nu
287 prop_binCRTT_electron2017_bin6
288 prop_binCRWS_electron2017_bin7_WrongSign
289 prop_binCRWS_muon2017_bin4_TVX
290 prop_binCRTT_muon2018_bin4
291 prop_binCRWS_muon2017_bin4_VG
292 prop_binCRTT_electron2017_bin5
293 prop_binCRWS_electron2017_bin7_TTTo2L2Nu
294 prop_binCRWS_electron2016M_bin6_WrongSign
295 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
296 prop_binCRTT_muon2016M_bin6
297 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
298 prop_binSR_muon2016M_bin8_WrongSign
299 prop_binCRWS_electron2017_bin6_WrongSign
300 prop_binCRTT_muon2017_bin7
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS1}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cS1} = 1.00^{+0.08}_{-0.43}$
301 prop_binCRWS_muon2017_bin6_VG
302 prop_binCRWS_electron2017_bin6_VG
303 prop_binCRWS_electron2017_bin6_Triboson
304 prop_binCRWS_muon2017_bin6_TTTo2L2Nu
305 prop_binCRWS_muon2017_bin6_WrongSign
306 lep_2017
307 prop_binCRWS_muon2017_bin6_TVX
308 prop_binCRWS_muon2017_bin6_Triboson
309 prop_binCRWS_muon2018_bin6_TVX
310 prop_binCRWS_electron2016M_bin8_Triboson
311 tau_vsmu_2017
312 prop_binCRWS_electron2016M_bin6_Triboson
313 prop_binCRWS_muon2018_bin6_ZZtoLep
314 prop_binCRF_muon2017_bin7_sm
315 prop_binCRWS_electron2016M_bin6_TVX
316 prop_binCRWS_electron2016M_bin6_ZZtoLep
317 prop_binCRF_electron2016M_bin6_sm
318 prop_binCRWS_electron2016M_bin8_ZZtoLep
319 prop_binCRWS_muon2017_bin6_ZZtoLep
320 tau_vsele_2017
321 prop_binCRF_electron2016M_bin7_sm
322 prop_binCRF_muon2016M_bin6_sm
323 prop_binCRF_muon2018_bin8_sm
324 prop_binCRF_electron2017_bin8_sm
325 prop_binCRF_muon2017_bin8_sm
326 prop_binCRWS_electron2017_bin7_TVX
327 prop_binCRWS_muon2017_bin4_ZZtoLep
328 VBS_2016M
329 VBS_2017
330 VBS_2018
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cS1}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cS1 = 1.00 +0.08 −0.43
331 prop_binCRF_electron2017_bin7_sm
332 prop_binCRF_muon2016M_bin8_sm
333 prop_binCRWS_electron2016M_bin8_sm
334 prop_binCRWS_electron2017_bin7_ZZtoLep
335 prop_binCRWS_electron2017_bin7_sm
336 prop_binCRWS_muon2018_bin6_sm
337 prop_binSR_muon2016M_bin8_sm
338 r
1.0 +0 −0
339 prop_binCRWS_electron2017_bin7_Triboson
340 tau_vsele_2018
341 lep_2018
342 tau_vsmu_2018
343 prop_binCRWS_electron2017_bin6_TVX
344 tau_vsmu_2016M
345 prop_binCRWS_muon2018_bin6_Triboson
346 tau_vsele_2016M
347 prop_binCRWS_muon2017_bin4_Triboson
348 lep_2016M
349 FES
350 prop_binCRWS_electron2017_bin8
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cS1